

those that are used extensively by people for economic gain. Increasingly, ecologists are looking at species preservation even in the context of cities. The field of **urban ecology** examines organisms and their environment in urban settings.

For the first time in history, more than half of the people on Earth live in cities. By the year 2030, 5 billion people are expected to be living in urban environments. As cities expand in number and size, protected areas that were once outside city boundaries become incorporated into urban landscapes. Ecologists are now studying cities as ecological laboratories, seeking to balance species preservation and other ecological needs with the needs of people.

One critical area of research centers on urban streams, including the quality and flow of their water and the organisms living in them. Urban streams tend to rise and fall more quickly after rain than natural streams. This rapid change in water level occurs because of the concrete and other impervious surfaces in cities as well as the drainage systems that route water out of cities as quickly as possible to avoid flooding. Urban streams also tend to have higher concentrations of nutrients and contaminants and are often straightened or even channeled underground.

Near Vancouver, Canada, ecologists and volunteers have worked to restore a degraded urban stream, Guichon Creek. They stabilized its banks, removed introduced plants, and planted native trees and shrubs along the creek (**Figure 56.22**). Their efforts returned the water flow and the communities of invertebrates and fish in the stream much closer to what they had been 50 years before the stream became degraded. A few years ago, ecologists successfully reestablished cutthroat trout in the stream. The trout are now thriving.

As cities continue to expand into the landscapes around them, understanding the ecological effects of this expansion will only increase in importance. Research on and conservation of urban habitats will continue to grow.

▼ **Figure 56.22** Volunteers working to remove invasive species along the urban Guichon Creek.



CONCEPT CHECK 56.3

1. What is a biodiversity hot spot?
2. How do zoned reserves provide economic incentives for long-term conservation of protected areas?
3. **WHAT IF?** Suppose a developer proposes to clear-cut a forest that serves as a corridor between two parks. To compensate, the developer also proposes to add the same area of forest to one of the parks. As a professional ecologist, how might you argue for retaining the corridor?

For suggested answers, see Appendix A.

CONCEPT 56.4

Earth is changing rapidly as a result of human actions

As we've discussed, landscape and regional conservation help protect habitats and preserve species. However, environmental changes that result from human activities are creating new challenges. As a consequence of human-caused climate change, for example, the place where a vulnerable species is found today may not be the same place that is needed for its preservation in the future. What would happen if *many* habitats on Earth changed so quickly that the locations of preserves today were unsuitable for their species in 10, 50, or 100 years? Such a scenario is increasingly possible.

The rest of this section describes four types of environmental change that humans are bringing about: nutrient enrichment, toxin accumulation, climate change, and ozone depletion. The impacts of these and other changes are evident not just in human-dominated ecosystems, such as cities and farms, but also in the most remote ecosystems on Earth.

Nutrient Enrichment

Human activity often removes nutrients from one part of the biosphere and adds them to another. Someone eating broccoli in Washington, DC, consumes nutrients that only days before were in the soil in California; a short time later, some of these nutrients will be in the Potomac River, having passed through the person's digestive system and a local sewage treatment facility. Likewise, nutrients in farm soil may run off into streams and lakes, depleting nutrients in one area, increasing them in another, and altering chemical cycles in both.

Farming illustrates how human activities can lead to nutrient enrichment. After vegetation is cleared from an area, the existing reserve of nutrients in the soil is depleted over time because many of these nutrients are exported from the area in crop biomass. How long this takes varies greatly. When some of the early North American prairie lands were first tilled, good crops could be produced for decades because the large store of organic materials in the soil continued to decompose and provide nutrients. By contrast, some cleared land in the tropics can be farmed for only one or two years because so little of the ecosystems'